

KRONOS Institutional Model Validation Pack

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1. Executive Summary

Phase 1 consolidates model-risk evidence for the KRONOS PD champion. The pack includes feature governance, calibration, proxy OOT stability testing, and challenger benchmarking. Final approval status: **AMBER**.

- OOT validation uses proxy record-order methodology because no true temporal field exists.

2. Champion Model Overview

Metric	Value
accuracy	0.8593
precision	0.7513
recall	0.6002
f1_score	0.6673
roc_auc	0.9068
model_health_score	0.8111
psi	0.0058
model_drift	NO DRIFT
train_auc	0.9534
overfitting_risk	MODERATE OVERFITTING RISK
feature_count	61

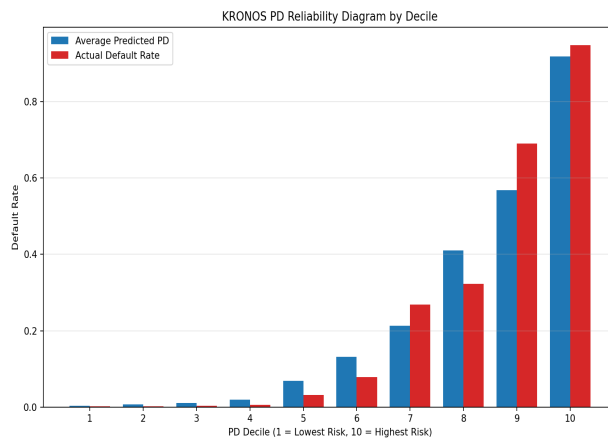
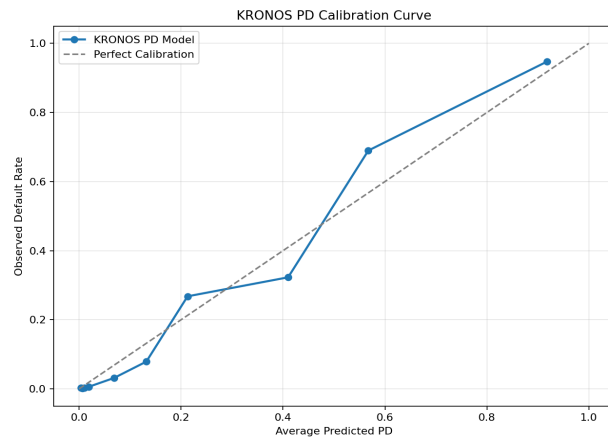
3. Feature Governance

Governance status: **PASSED**. Prohibited controls: borrower_id, customer_id, account_id, loan_id, application_id.

Identifier-like fields are excluded and training fails if borrower_id, customer_id, account_id, loan_id, or application_id enters final training features.

4. Calibration Assessment

Metric	Value
Calibration Status	PASS
Brier Score	0.080149
Predicted Default Rate	0.234826
Actual Default Rate	0.23514
Absolute Gap	0.000314
Max Decile Gap	0.122424



Decile	Obs	Default Rate	Avg PD	Gap
1.0	5000.0	0.0022	0.003752	0.001552
2.0	5000.0	0.0016	0.006769	0.005169
3.0	5000.0	0.0034	0.010205	0.006805
4.0	5000.0	0.006	0.01905	0.01305
5.0	5000.0	0.0316	0.068637	0.037037
6.0	5000.0	0.0788	0.131777	0.052977
7.0	5000.0	0.2678	0.2127	-0.0551
8.0	5000.0	0.3228	0.409949	0.087149
9.0	5000.0	0.6898	0.567376	-0.122424
10.0	5000.0	0.9474	0.91805	-0.02935

5. OOT Validation Assessment

Metric	Value
Methodology	Proxy OOT Validation
Split Method	chronological_record_ordering_fallback
True Temporal Field Found	False
OOT AUC	0.942631
OOT KS	0.739438
OOT F1	0.748771
PSI	0.001287
PSI Status	NO MATERIAL DRIFT

Limitation: Dataset does not contain origination, vintage, reporting, or observation dates. Validation uses chronological record-order fallback and should not be interpreted as true future-period model validation.

6. Challenger Model Assessment

Best challenger: **Logistic Regression**. AUC gap versus champion: 0.002287. Recommendation: Maintain Current VotingClassifier as champion. Challenger performance is broadly comparable and does not justify model replacement.

Model	Role	AUC	KS	F1	Brier
Logistic Regression	Challenger	0.909082	0.658498	0.675418	0.095236
Random Forest	Challenger	0.908391	0.656136	0.675801	0.095666
LightGBM	Challenger	0.906865	0.650967	0.665881	0.096578
Current VotingClassifier (Champion)	Champion	0.906795	0.653167	0.667297	0.096455
XGBoost	Challenger	0.905931	0.652736	0.668712	0.096845

7. Model Risk Assessment

Strengths

- Governed feature list excludes prohibited identifier fields.
- Calibration outputs include Brier score, decile analysis, and reliability charts.
- Proxy OOT framework quantifies score shift, PSI, and stability metrics.
- Challenger framework compares four challenger models against the champion.

Weaknesses

- Borrower-level dataset lacks true origination, vintage, reporting, or observation date.
- OOT validation is proxy record-order validation, not true future-period validation.
- LGD and EAD are outside this PD validation pack scope.
- Independent model validation sign-off remains simulated through local artifacts.

Assumptions

- Current VotingClassifier remains the approved champion by policy.

- Saved PD model, scaler, and feature list are the active production-demo artifacts.
- Merged credit dataset remains the validation population for Phase 1.

Limitations

- Dataset does not contain origination, vintage, reporting, or observation dates. Validation uses chronological record-order fallback and should not be interpreted as true future-period model validation.

8. Governance & Monitoring Assessment

Governance controls, calibration evidence, proxy OOT stability checks, and challenger benchmarking are now available. Continued monitoring should focus on PSI, calibration decile gaps, and introduction of true borrower-level temporal fields.

9. Recommended Actions

- Maintain VotingClassifier as champion.
- Do not replace the production-demo PD model based on challenger results.
- Add true borrower-level origination or observation date for future OOT validation.
- Continue monitoring calibration decile gaps and PSI trends.

10. Final Approval Status

AMBER